

Study Report: Prompt Engineering for Generative AI - Generative AI Engineer

Written by Gagan Pasupuleti

Family: Generative AI Engineer

Level: Beginner

Chapters: 7

Source: Reference: Prompt Engineering for Generative AI - James Phoenix & Mike Taylor

Official sources and free libraries

- <https://www.oreilly.com/library/view/prompt-engineering-for/9781098153434/>

Summary

Study report for *Prompt Engineering for Generative AI* by James Phoenix & Mike Taylor (Beginner level) mapped to the Generative AI Engineer role. Prompt patterns, few-shot, chain-of-thought, and evaluation.

Chapters

Track: Book-Intro

Chapter 1: About Prompt Engineering for Generative AI [Beginner]

Chapter focus: About Prompt Engineering for Generative AI** *(Level: Beginner)

This study report summarizes *Prompt Engineering for Generative AI* by James Phoenix & Mike Taylor for the Generative AI Engineer role. The resource is rated Beginner level. Prompt patterns, few-shot, chain-of-thought, and evaluation. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Prompt Engineering for Generative AI
# Author: James Phoenix & Mike Taylor
# Role: Generative AI Engineer
# Level: Beginner
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on Prompt Engineering for Generative AI.

When to use it

When learning Generative AI Engineer skills at Beginner level.

Where to use it

Generative AI Engineer training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Beginner]

Chapter focus: Why This Book Matters for Your Role** *(Level: Beginner)

Generative AI Engineer professionals use ideas from Prompt Engineering for Generative AI to solve real workplace problems. Prompt patterns, few-shot, chain-of-thought, and evaluation. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: Generative AI Engineer

Book focus: Prompt patterns, few-shot, chain-of-thought, and evaluation.

Recommended level: Beginner

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

Generative AI Engineer bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a beginner skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Beginner]

Chapter focus: Key Topics Covered** *(Level: Beginner)

The main topics in Prompt Engineering for Generative AI include practical concepts described as: Prompt patterns, few-shot, chain-of-thought, and evaluation. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Generative AI Engineer jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Generative AI Engineer Role Overview [Beginner]

Chapter focus: Generative AI Engineer Role Overview** *(Level: Beginner)

GenAI engineers build LLM-powered products: copilots, search, summarization, and codegen assistants. Skills span prompting, retrieval, evaluation, and safety - not just calling an API. 2025-2026 teams ship RAG before fine-tuning unless domain gap is proven.

Code Reference:

```
response = client.chat.completions.create(
    model='gpt-4o-mini',
    messages=[{'role':'system','content':'You are a helpful tutor.'},
               {'role':'user','content':'Explain SQL JOINS briefly.'}]
)
```

What it shows: System + user messages frame behavior - cheapest path to useful apps.

Topic guide

What it actually is

A GenAI engineer integrates LLMs into reliable user-facing applications.

When to use it

When unstructured knowledge or language tasks dominate the product.

Where to use it

Support bots, doc search, content generation, and code assistants.

Real use example

CodeQuest study copilot answers from official docs only - you own retrieval quality.

Chapter 5: Applied: LLM Fundamentals: Tokens and Context [Beginner]

Chapter focus: LLM Fundamentals: Tokens and Context** *(Level: Beginner)

Models read text as tokens (~4 chars English). Context window limits total input+output (128k+ on flagship 2025 models). Long prompts cost money and latency - summarize or retrieve instead of stuffing context.

Code Reference:

```
import tiktoken
enc = tiktoken.encoding_for_model('gpt-4o')
print(len(enc.encode('Hello, how many tokens am I?')))
```

What it shows: tiktoken estimates billable tokens before sending requests.

Topic guide

What it actually is

Tokenization and context limits shape every LLM application design.

When to use it

Sizing prompts, choosing models, and estimating cost.

Where to use it

All Chat Completions and embedding workloads.

Real use example

A 200-page PDF won't fit - chunking + RAG is mandatory not optional.

Chapter 6: Applied: Prompt Engineering Patterns [Beginner]

Chapter focus: Prompt Engineering Patterns** *(Level: Beginner)

Patterns: zero-shot, few-shot, chain-of-thought, JSON mode, structured outputs (2024+). Be explicit about format, length, and refusal conditions. Version prompts in Git.

Code Reference:

```
SYSTEM = """Answer using only provided context. If unsure, say I don't know.  
Respond as JSON: {"answer": str, "citations": [int]}"""
```

What it shows: JSON schema reduces parsing bugs vs free-form answers.

Topic guide

What it actually is

Prompt engineering steers model behavior without weight updates.

When to use it

Every LLM feature before investing in fine-tuning.

Where to use it

Extraction, classification, tutoring, and support macros.

Real use example

Few-shot examples fix date format inconsistencies in invoice parsing.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Beginner]

Chapter focus: Study Plan and Practice** *(Level: Beginner)

Finish Prompt Engineering for Generative AI with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Generative AI Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes  
Week 2: Exercises  
Week 3: Mini project  
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a beginner project screenshot.